



Stage A · Construction (yr 0)

Inputs: raw materials · diesel · electricity

A1-A3 Materials & manufacturing

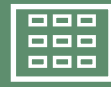
- HDPE loops; grout
- Heat pumps; pumps
- Working fluid; steel

A4-A5 Transport & installation

- Freight to site
- Drilling & grouting (115)
- Trenching; pumphouse



Outputs: construction waste · emissions



Stage B · Operation (yr 1-50)

Inputs: spare parts · grid electricity

B2-B4 Maintenance & replacement

- Maintenance; top-ups
- Pump swaps (15/30/45 yr)
- Heat-pump swap (yr 25)

B6 Operational energy

- Grid electricity (no combustion)
- Building heat pumps; pumps
- Heating, cooling & DHW



Outputs: replaced parts · grid emissions



Stage C · End of life (yr 50)

Inputs: energy for decommissioning

C1-C2 Decommissioning & transport

- Well sealing; removal
- Transport to disposal

C3-C4 Treatment & disposal

- Scrap credit; HDPE landfill
- Refrigerant/solvent treatment
- Boreholes left in place



Outputs: recovered scrap · final waste